plants and animals at home in a particular habitat.” From this, the idea of the ecosystem as a “superorganism” later developed.

“For Clements, climates are like genomes, and vegetation is like an organism whose characteristics its genome determines.”

Christopher Elliott

Philosopher of science

A fluctuating process

Clements’s ideas were challenged from the start, although the idea of a “steady state” proved influential and dominated thinking about ecosystems up until the 1960s. However, scientists realized that communities change constantly in response to conditions, and it is almost impossible to observe a true climax community. A $10,000 prize for identifying such a community, offered by American botanist Frank Egler in the 1950s, was never claimed. Despite the difficulties, ecologists continued to use the theory of a climax community to decide how to respond to invasive species that threatened to disturb an established native community, and in recent decades Clements’s ideas have regained support.

Succession remains a core principle of ecology. In general, early succession phases consist of fast-growing and well-dispersed species that are replaced by more competitive species. Initially, ecologists thought that ecological succession ended in what they described as the climax phase, when the ecosystem reached a stable equilibrium. However, it is now accepted that ecological succession is a dynamic process that is constantly in flux.